Power monoids and their arithmetic

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Der Wissenschaftsfonds.

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Monoids, units and atoms

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Set $\mathcal{A}(M) =$

$$\{x \in M \setminus M^\times \mid \text{ for all } y, z \in M, x = y + z \Rightarrow y \in M^\times \text{ or } z \in M^\times\},$$

called the **set of atoms** of M.

Length sets

For each $a \in M$, let

$$\mathsf{L}(a) = \{n \in \mathbb{N}_0 \mid a = \sum_{i=1}^{n} u_i \text{ for some atoms } (u_i)_{i=1}^{n} \text{ of } M\},$$

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Let

$$\mathcal{L}(M) = \{\mathsf{L}(a) \mid a \in M, \mathsf{L}(a) \neq \emptyset\},$$

called the **system of length sets** of M.

Elasticity and atomicity

For each $a \in M$, let

$$\rho(a) = \begin{cases} \sup(\mathsf{L}(a))/\min(\mathsf{L}(a)) & \text{if } \min(\mathsf{L}(a)) > 0 \\ 1 & \text{else} \end{cases},$$

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We say that M is **atomic** if every nonunit of M is a finite sum of atoms.

Reduced monoids and the factorization monoid

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Let $\pi : \mathsf{Z}(M) \to M$ be the **factorization homomorphism** defined by $\pi(\prod_{i=1}^n a_i) = \sum_{i=1}^n a_i$ for all $n \in \mathbb{N}_0$ and atoms $(a_i)_{i=1}^n$ of M.

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For each $a \in M$, set

$$\mathsf{Z}(a) = \{z \in \mathsf{Z}(M) \mid \pi(z) = a\},$$

called the **set of factorizations** of a.

Relatively prime and (relatively) cancellative elements

If $a, b \in M$, then we say that a and b are **relatively prime** if for all $c, d, e \in M$ with $a = c + d$ and $b = c + e$, it follows that $c \in M^\times$.

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We say that M is **cancellative** if every element of M is cancellative.

Moreover, M is called **unit cancellative** if for all $b, c \in M$ with $b = b + c$, it follows that $c \in M^\times$.

On the cancellativity conditions

- Units are cancellative and cancellative elements are relatively cancellative.
- If M is reduced, then each atom of M is relatively cancellative.
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Let R be an integral domain and $\mathcal{I}$ be the monoid of nonzero ideals of R.

- Every cancellative monoid is unit cancellative.
- If R is noetherian, then $\mathcal{I}$ is unit cancellative.
- $\mathcal{I}$ is cancellative if and only if R is an almost Dedekind domain.

BF-/half-factorial/fully elastic monoids

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Note that if

$$\mathcal{L}(M) = \{\{0\}, \{1\}\} \cup \{A \subseteq \mathbb{N}_0 \mid \emptyset \neq A \text{ is finite, } \min(A) \geq 2\},$$

then M is fully elastic.

Power monoids

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Let $+ : \mathcal{P}_{\text{fin}}(M) \times \mathcal{P}_{\text{fin}}(M) \to \mathcal{P}_{\text{fin}}(M)$ be defined by $A + B = \{a + b \mid a \in A, b \in B\}$ for each $A, B \in \mathcal{P}_{\text{fin}}(M)$.

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Besides that $\mathcal{P}_{\text{fin},0}(M)$ is a submonoid of $\mathcal{P}_{\text{fin}}(M)$, called the **reduced finitary power monoid** of M.

Some properties of power monoids

Let $\mathcal{H} = \mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0)$ and let $a, b \in \mathbb{N}$ be such that $a < b$.

- $\mathcal{H}^\times = \{\{0\}\}$ is the set of all cancellative elements of $\mathcal{H}$.
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Let $M = (\mathbb{N}, \cdot)$. Then the set of prime numbers $\mathbb{P}$ is the set of atoms of M and $\lim_{n\to\infty} \frac{|\{m\in\mathbb{P} | m\leq n\}|}{n} = 0$.

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For each $n \in \mathbb{N}$, let $\mathcal{A}_n = \{A \in \mathcal{A}(\mathcal{H}) \mid \max(A) \le n\}$ and let $\mathcal{P}_n = \{A \in \mathcal{H} \mid \max(A) \le n\}$. Then $\lim_{n\to\infty} \frac{|\mathcal{A}_n|}{|\mathcal{P}_n|} = 1$.

Importance of power monoids, part I

Let H and D be monoids. It is clear that if H and D are isomorphic, then $\mathcal{P}_{\mathrm{fin}}(H)$ and $\mathcal{P}_{\mathrm{fin}}(D)$ are isomorphic (and $\mathcal{P}_{\mathrm{fin},0}(H)$ and $\mathcal{P}_{\mathrm{fin},0}(D)$ are isomorphic, too).

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Power monoids were introduced and studied by Tamura and Shafer in 1967. Under what circumstances is the converse to the aforementioned facts true?

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Power monoids were introduced and studied by Tamura and Shafer in 1967. Under what circumstances is the converse to the aforementioned facts true?

This is a recent research topic and there are many partial results by Bienvenu, Geroldinger, Rago, Tringali and others.

Importance of power monoids, part II

Let H be a monoid and let $\mathrm{Aut}(H)$ be the automorphism group of H (i.e., the set of all monoid automorphism on H under composition). How does $\mathrm{Aut}(H)$ look like?

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Theorem (Tringali, Yan, 2025)

Let $\mathcal{H} = \mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0)$. Then $\mathrm{Aut}(\mathcal{H}) = \{\mathrm{id}, \sigma\}$, where id is the identity map and $\sigma : \mathcal{H} \to \mathcal{H}$ is defined by $\sigma(X) = \{\max(X) - x \mid x \in X\}$ for each $X \in \mathcal{H}$.

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For more results on this topic see, for instance, the papers of Rago, Tringali, Wen.

Importance of power monoids, part III

Let M be an additively written monoid and let $\mathcal{H} \in \{\mathcal{P}_{\text{fin}}(M), \mathcal{P}_{\text{fin},0}(M)\}$. What can be said about the arithmetic of $\mathcal{H}$? This includes (but is not limited to) the following problems.

Importance of power monoids, part III

Let M be an additively written monoid and let $\mathcal{H} \in \{\mathcal{P}_{\mathrm{fin}}(M), \mathcal{P}_{\mathrm{fin},0}(M)\}$. What can be said about the arithmetic of $\mathcal{H}$? This includes (but is not limited to) the following problems.

- Characterize when $\mathcal{H}$ is atomic/a BF-monoid/a half-factorial monoid.
- Determine $\rho(\mathcal{H})$, the elasticity of $\mathcal{H}$.
- Describe $\mathcal{L}(\mathcal{H})$ and characterize when $\mathcal{H}$ is fully elastic.

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- Describe $\mathcal{L}(\mathcal{H})$ and characterize when $\mathcal{H}$ is fully elastic.

More recent results were established by Aggarwal, Cossu, Gotti, Lu, Tringali and others.

Motivational results, part I

Theorem (Chapman, McClain, 2005)

Let $D = \{f \in \mathbb{Q}[X] \mid f(\mathbb{Z}) \subseteq \mathbb{Z}\}$ be the ring of integer-valued polynomials and let $H = D \setminus \{0\}$.
Then H is fully elastic.

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Theorem (Frisch, 2013)

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Krull monoids

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$H_{\mathrm{red}} = \{xH^\times \mid x \in H\}$ is the **reduced monoid associated to** H.

Moreover, H is called a **Krull monoid** if we can find a free abelian monoid F such that H_{red} is a submonoid of F such that for all $a, b \in H_{\mathrm{red}}$ and $c \in F$ with $a = bc$, it follows that $c \in H_{\mathrm{red}}$.

Monoids of (plus-minus weighted) zero-sum sequences

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Let $\mathcal{B}(G_0) = \{\prod_{i=1}^{n} g_i \in \mathcal{F}(G_0) \mid \sum_{i=1}^{n} g_i = 0\}$, called the **monoid of zero-sum sequences** of G_0 (or called the **block monoid** of G_0).

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Let

$$\mathcal{B}_{\pm}(G_0) = \{\prod_{i=1}^{n} g_i \in \mathcal{F}(G_0) \mid \sum_{i=1}^{n} \varepsilon_i g_i = 0 \text{ for some } \varepsilon_i \in \{-1, 1\}\},$$

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Properties of the aforementioned monoids

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Then $\mathcal{B}(G_0)$ is a Krull monoid and if H is a Krull monoid, then there exists an abelian group G and a subset $G_0 \subseteq G$ such that H_{red} is isomorphic to $\mathcal{B}(G_0)$.

Properties of the aforementioned monoids

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Then $\mathcal{B}(G_0)$ is a Krull monoid and if H is a Krull monoid, then there exists an abelian group G and a subset $G_0 \subseteq G$ such that H_{red} is isomorphic to $\mathcal{B}(G_0)$.

Also note that $\mathcal{B}_{\pm}(G)$ is a Krull monoid if and only if G is an elementary 2-group.

Motivational results, part II

Theorem (Kainrath, 1999)

Let H be a Krull monoid and let G be an infinite abelian group such that $H_{\mathrm{red}} \cong \mathcal{B}(G)$.
Then $\mathcal{L}(H) = \{\{0\}, \{1\}\} \cup \{A \subseteq \mathbb{N}_0 \mid \emptyset \neq A \text{ is finite, } \min(A) \geq 2\}$.

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Theorem (Geroldinger, Kainrath, 2026)

Let G be an infinite abelian group and let $H = \mathcal{B}_{\pm}(G)$ be the monoid of plus-minus weighted zero-sum sequences of G.
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A conjecture and some partial results

Conjecture (Fan, Tringali, 2018)

$\mathcal{L}(\mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0)) = \{\{0\},\{1\}\} \cup \{A \subseteq \mathbb{N}_0 \mid \emptyset \neq A \text{ finite}, \min(A) \geq 2\}$.

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$\{[2,n],\{2,n\},\{n\} \mid n \in \mathbb{N}_{\geq 2}\} \subseteq \mathcal{L}(\mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0))$.

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Corollary

$\mathcal{P}_{\text{fin}}(\mathbb{N}_0)$ is fully elastic and $\rho(\mathcal{P}_{\text{fin}}(\mathbb{N}_0)) = \rho(\mathcal{P}_{\text{fin},0}(\mathbb{N}_0)) = \infty$.

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Corollary

$\mathcal{P}_{\mathrm{fin}}(\mathbb{N}_0)$ is fully elastic and $\rho(\mathcal{P}_{\mathrm{fin}}(\mathbb{N}_0)) = \rho(\mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0)) = \infty$.

Theorem (Antoniou, 2020)

$\{[2,n] \cup \{n+m\} \mid n,m \in \mathbb{N}_{\geq 2}\} \subseteq \mathcal{L}(\mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0))$.

Some complementary results

Theorem (Geroldinger, Halter-Koch, 2006)

Let G be an abelian group and let $G_0 \subseteq G$ be a finite subset. Then $\mathcal{L}(\mathcal{B}(G_0)) \subsetneq \{\{0\}, \{1\}\} \cup \{A \subseteq \mathbb{N}_0 \mid \emptyset \neq A \text{ is finite, } \min(A) \geq 2\}$. (More is true: In fact $\mathcal{B}(G_0)$ satisfies the structure theorem for sets of lengths.)

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Theorem (Baginski, Chapman, Crutchfield, Kennedy, Wright, 2006)

Let K be a field, let $S \neq \mathbb{N}_0$ be a numerical semigroup (i.e., S is an additive submonoid of $\mathbb{N}_0$ such that $\mathbb{N}_0 \setminus S$ is finite) and let $H = \{f \in K[\![X]\!] \setminus \{0\} \mid f_i = 0 \text{ for all } i \in \mathbb{N}_0 \setminus S\}$. Then H is not fully elastic.

Why relatively cancellative elements are useful

Proposition (R., 2025)

Let $\mathcal{H} = \mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0)$ and let $X, Y \in \mathcal{H}$ be relatively cancellative such that $\gcd(Y) > 2\max(X)$.
Then $X + Y$ is relatively cancellative, $\mathsf{Z}(X+Y) = \mathsf{Z}(X)\mathsf{Z}(Y)$ and $\mathsf{L}(X+Y) = \mathsf{L}(X) + \mathsf{L}(Y)$.

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Corollary

Let $\mathcal{H} = \mathcal{P}_{\text{fin},0}(\mathbb{N}_0)$ and let $X, Y \in \mathcal{H}$ be relatively cancellative.
Then there is some relatively cancellative $Z \in \mathcal{H}$ such that $\mathsf{L}(X) + \mathsf{L}(Y) = \mathsf{L}(Z)$.

Some counterexamples

Examples

Let $A = \{0, 1, 2, 3\}$, $B = \{0, 7\}$, $C = \{0, 1\}$, $D = \{0, 3, 6, 9\}$, $E = \{0, 2\}$.

(1) A and D are not relatively cancellative and B, C and E are relatively cancellative.

(2) $\gcd(B) > 2 \max(A)$, $\gcd(D) > 2 \max(C)$ and $\gcd(E) = 2 \max(C)$.

(3) $\mathsf{L}(A + B) = \{2, 3, 4\} \supsetneq \{3, 4\} = \mathsf{L}(A) + \mathsf{L}(B)$.

(4) $\mathsf{L}(C + D) = \{2, 3, 4\} \supsetneq \{3, 4\} = \mathsf{L}(C) + \mathsf{L}(D)$.

(5) $\mathsf{L}(C + E) = \{2, 3\} \supsetneq \{2\} = \mathsf{L}(C) + \mathsf{L}(E)$.

How to construct nontrivial relatively cancellative elements

Theorem (R., 2025)

Let $\mathcal{H} = \mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0)$ and let $(n_j)_{j=0}^{\infty}$, $(A_j)_{j=0}^{\infty}$, $(B_j)_{j=0}^{\infty}$, $(C_j)_{j=0}^{\infty}$, $(D_j)_{j=0}^{\infty}$ and $(S_j)_{j=0}^{\infty}$ be defined recursively as follows.
Set $n_0 = 0$, $A_0 = \{0,1\}$, $B_0 = \{0\}$,
$C_0 = \{0\}$, $D_0 = \{0,1\}$ and $S_0 = \{0,1\}$.
For each $i \in \mathbb{N}_0$, let $n_{i+1} \in \mathbb{N}$ with $n_{i+1} \geq 3\max(D_i)$,
$A_{i+1} = A_i \cup \{1 + n_{i+1}\}$, $B_{i+1} = B_i \cup (n_{i+1} + D_i)$,
$C_{i+1} = C_i \cup \{n_{i+1}\}$, $D_{i+1} = D_i + \{0, 1 + n_{i+1}\}$ and
$S_{i+1} = C_{i+1} + D_{i+1}$.
Then for each $i \in \mathbb{N}$, $S_i \in \mathcal{H}$ is relatively cancellative,
$\mathsf{Z}(S_i) = \{A_i B_i, C_i \prod_{j=0}^{i}\{0, 1+n_j\}\}$ and $\mathsf{L}(S_i) = \{2, i+2\}$.

Sketch of the proof I

Step 1: Show by induction that for each $i \in \mathbb{N}$, $A_i, B_i, C_i \in \mathcal{A}(\mathcal{H})$, $D_i \in \mathcal{H}$, $\mathsf{Z}(D_i) = \{\prod_{j=0}^{i}\{0, 1+n_j\}\}$, $\mathsf{L}(D_i) = \{i+1\}$, $\max(A_i \cup B_i \cup C_i) < \max(D_i)$, $A_i = \{0\} \cup (1 + C_i)$ and $A_i + B_i = C_i + D_i$.

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Step 2: Prove that for each $i \in \mathbb{N}$,
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Step 2a: Show that $\mathsf{Z}(S_1) = \{A_1 B_1, C_1 \prod_{j=0}^{1}\{0, 1+n_j\}\}$ (straightforward) and observe that $\{A_i B_i, C_i \prod_{j=0}^{i}\{0, 1+n_j\}\} \subseteq \mathsf{Z}(S_i)$ for each $i \in \mathbb{N}$ by Step 1.

Sketch of the proof II

Step 2b: Let $i \in \mathbb{N}$ be such that $\mathsf{Z}(S_i) \subseteq \{A_i B_i, C_i \prod_{j=0}^{i}\{0, 1+n_j\}\}$. Prove that $\mathsf{Z}(S_{i+1}) \subseteq \{A_{i+1} B_{i+1}, C_{i+1} \prod_{j=0}^{i+1}\{0, 1+n_j\}\}$.

Sketch of the proof II

Step 2b: Let $i \in \mathbb{N}$ be such that $\mathsf{Z}(S_i) \subseteq \{A_i B_i, C_i \prod_{j=0}^{i} \{0, 1 + n_j\}\}$. Prove that $\mathsf{Z}(S_{i+1}) \subseteq \{A_{i+1} B_{i+1}, C_{i+1} \prod_{j=0}^{i+1} \{0, 1 + n_j\}\}$.

Step 2b(i): Let $z \in \mathsf{Z}(S_{i+1})$. Then $z = \prod_{i=1}^{n+1} Y_i$ with $n \in \mathbb{N}$ and $(Y_i)_{i=1}^{n+1}$ atoms of $\mathcal{H}$. There is some $j \in [1, n+1]$ such that $n_1 \in Y_j$. Without restriction, let $n_1 \in Y_{n+1}$. Set $X = \sum_{i=1}^{n} Y_i$ and $Y = Y_{n+1}$. Then $S_{i+1} = X + Y$, $C_{i+1} \subseteq Y$ and $X \subseteq D_{i+1}$.

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Set $X' = X \cap [0, n_{i+1} - 1]$ and $Y' = Y \cap [0, n_{i+1} - 1]$. Show that $S_i = X' + Y'$.

Conclude that either (1) $X' = A_i$ and $Y' = B_i$ or (2) there is some $W \subseteq [0, i]$ such that $X' = \{\sum_{j \in E}(1 + n_j) \mid E \subseteq W\}$ and $Y' = \{\sum_{j \in E}(1 + n_j) \mid E \subseteq [0, i] \setminus W\} + C_i$.

Sketch of the proof III

Step 2b(ii): First let $X' = A_i$ and $Y' = B_i$.
Prove that $X = A_{i+1}$ and $Y = B_{i+1}$.
Then $z = A_{i+1}B_{i+1}$.

Sketch of the proof III

Step 2b(ii): First let $X' = A_i$ and $Y' = B_i$.
Prove that $X = A_{i+1}$ and $Y = B_{i+1}$.
Then $z = A_{i+1}B_{i+1}$.

Next let $W \subseteq [0, i]$ be such that $X' = \{\sum_{j \in E}(1 + n_j) \mid E \subseteq W\}$ and $Y' = \{\sum_{j \in E}(1 + n_j) \mid E \subseteq [0, i] \setminus W\} + C_i$.
Show that $W = [0, i]$, $X = \{\sum_{j \in E}(1 + n_j) \mid E \subseteq [0, i + 1]\}$ and $Y = C_{i+1}$.
Then $z = C_{i+1} \prod_{j=0}^{i+1}\{0, 1 + n_j\}$.

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Show that $W = [0, i]$, $X = \{\sum_{j \in E}(1 + n_j) \mid E \subseteq [0, i + 1]\}$ and $Y = C_{i+1}$.
Then $z = C_{i+1} \prod_{j=0}^{i+1}\{0, 1 + n_j\}$.

Step 3: Let $i \in \mathbb{N}$. Conclude that $S_i \in \mathcal{H}$ is relatively cancellative and $\mathsf{L}(S_i) = \{2, i + 2\}$. (The former follows from the fact that for all $u, v \in \mathsf{Z}(S_i)$ with $\gcd(u, v) \neq 1$, it follows that $u = v$ and the latter is obvious.)

A few consequences

Corollary

Let $\mathcal{U} \subseteq \mathcal{P}_{\mathrm{fin}}(\mathbb{N}_0)$ be the submonoid generated by $\{\{1\}, \{2, n\} \mid n \in \mathbb{N}_{\geq 3}\}$. Then $\mathcal{U} \subseteq \mathcal{L}(\mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0))$.

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Corollary

If $L \subseteq \mathbb{N}_{\geq 2}$ is a nonempty finite arithmetical progression of length n and $\min(L) \geq 2n$, then $L \in \mathcal{L}(\mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0))$.

A complementary result

Proposition (R., 2025)

Let $\mathcal{H} = \mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0)$ and let $X \in \mathcal{H}$ and $n \in \mathbb{N}$ be such that $n > 2\max(X)$.
Set $\mathcal{M} =$

$$\{(A, B, C) \in \mathcal{H}^3 \mid A + B = A + C = X, B \text{ and } C \text{ are rel. prime}\}.$$

Set $\mathcal{N} = \{A \in \mathcal{H} \mid \text{there are } B, C \in \mathcal{H} \text{ such that } (A, B, C) \in \mathcal{M}\}$.
Then $\mathsf{Z}(X + \{0, n\}) = \bigcup_{(A,B,C)\in\mathcal{M}} (B \cup (n + C))\mathsf{Z}(A)$ and $\mathsf{L}(X + \{0, n\}) = 1 + \bigcup_{A\in\mathcal{N}} \mathsf{L}(A)$.

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Corollary

$[k, k + 2] \in \mathcal{L}(\mathcal{P}_{\mathrm{fin},0}(\mathbb{N}_0))$ for each $k \in \mathbb{N}_{\geq 2}$.

Examples I

Example

Let $X = \{0, 1, 2, 4, 5, 6, 10, 11, 12, 14, 15, 16, 20, 21, 22, 24, 25, 26\}$.

$$\begin{aligned} X =& \{0, 1\} + \{0, 1\} + \{0, 4\} + \{0, 10\} + \{0, 10\} \\ =& \{0, 4\} + \{0, 10\} + \{0, 1, 2, 6, 10, 11, 12\} \\ =& \{0, 4\} + \{0, 1, 2, 6, 10, 11, 12, 20, 21, 22\}. \end{aligned}$$

X is not relatively cancellative and $\mathsf{L}(X) = \{2, 3, 5\}$.

Examples II

Example

Let $X = \{0, 1, 3, 4, 5, 7, 8, 13, 14, 16, 17, 18, 20, 21, 22, 24, 25\} \cup \{30, 31, 33, 34, 35, 37, 38\}$.

$$\begin{aligned} X &= \{0, 1\} + \{0, 3\} + \{0, 4\} + \{0, 13\} + \{0, 17\} \\ &= \{0, 1, 4\} + \{0, 3, 4\} + \{0, 13\} + \{0, 17\} \\ &= \{0, 4\} + \{0, 1, 3, 4, 13, 14, 16, 18, 20, 21, 30, 31, 33, 34\}. \end{aligned}$$

X is not relatively cancellative and $\mathsf{L}(X) = \{2, 4, 5\}$.

Examples III

Example

Let $X = \{0, 1, 4, 5, 10, 11, 12, 14, 15, 16, 19, 20, 21, 22, 25, 26, 29\} \cup \{30, 42, 43, 46, 47, 52, 53, 54, 56, 57, 58, 61, 62, 63, 64, 67, 68, 71, 72\}$.

$$\begin{aligned} X =& \{0,1\} + \{0,4\} + \{0,10\} + \{0,11,15\} + \{0,42\} \\ =& \{0,1\} + \{0,10\} + \{0,4,10,11,15,19\} + \{0,42\} \\ =& \{0,10,11\} + \{0,1,4,5,10,11,15,19\} + \{0,42\}. \end{aligned}$$

X is not relatively cancellative and $\mathsf{L}(X) = \{3, 4, 5\}$.

Another application

Let M be an additively written monoid. We say that M is **nontorsion** if $\{na \mid n \in \mathbb{N}\}$ is infinite for some $a \in M$.

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Theorem (R., 2025)

Let G be an abelian group, let M be an atomic nontorsion submonoid of G and let $\mathcal{H} \in \{\mathcal{P}_{\mathrm{fin}}(M), \mathcal{P}_{\mathrm{fin},0}(M)\}$.

(1) The submonoid of $\mathcal{P}_{\mathrm{fin}}(\mathbb{N}_0)$ generated by $\{\{1\}, \{2, n\} \mid n \in \mathbb{N}_{\geq 3}\}$ is contained in $\mathcal{L}(\mathcal{H})$.

(2) $\{[k, k+2] \mid k \in \mathbb{N}_{\geq 2}\} \subseteq \mathcal{L}(\mathcal{H})$.

(3) $\mathcal{H}$ is fully elastic.

Open problems

Let $\mathcal{H} = \mathcal{P}_{\text{fin},0}(\mathbb{N}_0)$.

(1) Is $\mathcal{L}(\mathcal{H}) = \{\mathsf{L}(A) \mid A \in \mathcal{H} \text{ is relatively cancellative}\}$? In particular, is there some relatively cancellative $X \in \mathcal{H}$ such that $\mathsf{L}(X) = \{2, 3, 4\}$ or $\mathsf{L}(X) = \{3, 4, 5\}$

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(3) Is the conjecture of Fan and Tringali true?

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Thank you for your attention!